



Reducing contextual noise in review-based recommendation via aspect term extraction and attention modeling

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ABSTRACT

Information and Communication Technology (ICT) advances have accelerated e-commerce growth, exposing users to an overwhelming number of products and producing information overload. Recommender systems mitigate this problem by modeling user-item interactions, but traditional matrix factorization (MF) methods suffer from data sparsity. Prior work leverages user-generated review text to supply semantic cues about preferences and product attributes. However, most methods process whole reviews indiscriminately, mixing aspect-relevant content with contextual noise. This noise reduces the informational density, defined as the proportion of aspect-relevant content in a document, thereby degrading textual representations and model accuracy. We propose the Aspect Term-aware Recommender System (ATRS), which incorporates aspect-level semantics into review-based recommendation. ATRS applies a Bidirectional Encoder Representations from Transformers (BERT)-based aspect term extraction (ATE) model to identify product-related terms and filter irrelevant content, increasing informational density. Extracted aspect terms are encoded by a convolutional neural network (CNN) and aggregated with self-attention to produce aspect-aware user and item representations. Experiments on Amazon and Yelp datasets show ATRS outperforms baselines, producing average improvements of 19.54% in mean absolute error (MAE) and 11.89% in root mean square error (RMSE). Results confirm the benefit of aspect-level refinement and optimizing informational density in review-based recommender systems.

1. Introduction

Information and communication technology (ICT) advances have accelerated growth in the e-commerce market, exposing consumers to an even larger set of product choices. Although more options broaden choice, they also make it harder for users to find suitable products, producing information overload [1]. Recommender systems address this problem by providing personalized item suggestions based on users' historical behavior [2]. Matrix factorization (MF) is a representative approach that maps users and items into a latent factor space to model interactions [3,4]. Because MF relies solely on rating data, it remains vulnerable to data sparsity [1,5].

To mitigate sparsity, review-based recommender systems that incorporate user-written reviews have been widely studied [2].

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Reviews contain explicit evaluations and richer cues about user preferences and item attributes than ratings alone [6,7]. Many systems aggregate all reviews for a user or an item into a single document and derive representations from the consolidated text. For example, DeepCoNN uses two parallel convolutional neural network (CNN) to encode user and item review documents independently [2]. NARRE introduces an attention mechanism to capture the relative importance of individual reviews [8]. AENAR combines CNN-based review encodings with attention to further improve performance [6].

These approaches have a common limitation: they typically use the full review content. In practice, reviews mix aspect-related information about item attributes with contextual noise such as personal experiences or situational details [9,10]. From an information-theoretic perspective, including such noise significantly reduces informational density—the amount of meaningful information per text unit—which degrades representation quality and predictive performance [11,12]. For example, the review in Fig. 1 contains product-related terms such as “tone knob,” “fretboard,” and “ringing tone,” while the rest of the text is contextual noise that is irrelevant to the recommendation. Using the entire review reduces the signal-to-noise ratio and impairs learning efficiency [13,14].

Prior attempts to reduce noise include stop-word removal, TF-IDF weighting, and attention mechanisms [15,16]. These methods rely on frequency-based signals or reweight tokens, but they do not remove irrelevant tokens outright; semantic importance can therefore remain obscured [17]. Topic-modeling or frequency-based rules to Aspect Term Extraction (ATE) have also been explored [7,18], yet they often depend on domain-specific knowledge and show limited generalization performance [5,8]. Aspect-based sentiment analysis (ABSA) has been leveraged to capture fine-grained user preferences [19], but many ABSA-based methods use sentiment polarity or aspect-category attention rather than explicitly extracting aspect terms, which limits their ability to filter out irrelevant content.

Although large pretrained language models have driven significant advances in semantic understanding, review-based recommendation has only partially exploited these capabilities [20]. In particular, methods that selectively extract product-related information from reviews using pretrained language models and feed that information directly into recommendation architectures are still underexplored [17,21].

To fill this research gap, we propose the Aspect Term-aware Recommender System (ATRS) framework. ATRS improves preference prediction by extracting key aspect terms from reviews and building aspect-aware representations of users and items. Concretely, ATRS (1) uses a pretrained ATE model built on Bidirectional Encoder Representations from Transformers (BERT) to accurately extract aspect terms and increase informational density. (2) transforms extracted aspect terms into semantic vectors via a CNN-based architecture, and (3) integrates those vectors into user and item representations with a self-attention mechanism for preference predictions. We validate ATRS on large-scale Amazon and Yelp datasets. Our main contributions of this study are as follows.

- We introduce the concept of informational density into review-based recommender systems and apply it systematically. By reducing contextual noise and increasing informational density, ATRS improves predictive performance over existing review-based models.
- We present ATRS, a hybrid framework that integrates a pretrained BERT-based ATE model with CNN and self-attention modules to construct aspect-aware user and item representations.
- We evaluate ATRS on large-scale Amazon and Yelp datasets across e-commerce and restaurant domains. Ablation analyses confirm the contribution of each component and the method's generalizability.

The remainder of the paper is organized as follows. Section 2 reviews related work on review-based recommendation and ATE. Section 3 describes the ATRS architecture. Section 4 presents datasets, experiments, and results. Section 5 discusses theoretical and practical implications. Section 6 concludes and outlines directions for future research.

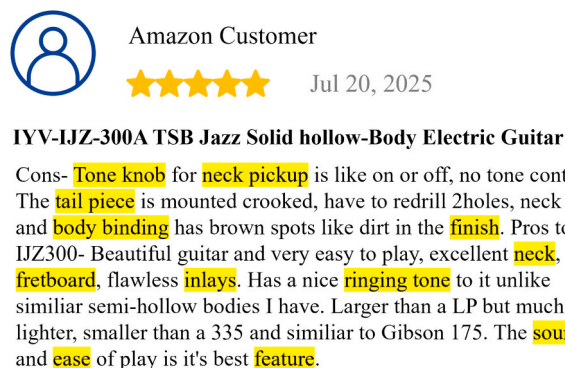


Fig. 1. Example of an Online Review.

2. Related works

2.1. Review-based recommender systems

With the rapid increase of online content, consumers face information overload, which makes accurate decision-making difficult [1]. To address this challenge, recommender systems analyze users' interaction histories to provide personalized options aligned with their preferences [2]. Early recommender-system studies focused on latent factor models based on ratings, such as MF [3] and neural collaborative filtering (NCF) [4]. However, these models suffer from limited predictive performance due to data sparsity [1,5]. In response, user-generated review texts, which explicitly convey reasons for evaluations and user preferences, have attracted increasing attention [7].

Review-based recommendation primarily leverages two document types: user documents that aggregate all reviews written by a user, and item documents that aggregate reviews from multiple consumers. These documents capture user preferences and item characteristics [2,6]. McAuley and Leskovec [7] proposed HFT, which jointly learns latent topics from review documents and rating factors to address the cold start problem. Topic models based on Latent Dirichlet Allocation (LDA), however, rely on word co-occurrence statistics and are limited in capturing contextual semantics or linguistic diversity [5,22].

To overcome these limitations, later studies employed neural architectures to learn richer contextual and semantic representations. For example, Zheng, Noroozi and Yu [2] proposed DeepCoNN, a model that uses a CNN encoder to address the limitations of bag-of-words representations and to learn contextual representations of user and item review documents. More recently, Lim, Li, Yang and Kim [23] proposed MFNR, which integrates embeddings from different pretrained language models, such as BERT [24] and RoBERTa [25] to enhance document representations.

Other studies emphasized that not all review content is equally essential and propose approaches to highlight key aspects. Chen, Zhang, Liu and Ma [8] introduced NARRE, which assigns greater weights to reviews that strongly reflect user preferences via review-level attention. Zhang, Sun, Cheng and Dong [6] proposed AENAR, which employs aspect-level attention to capture variations in user aspect emphasis. Ma and Liu [26] proposed SAFMR, which applies self-attention to capture the relative importance of different features within reviews.

In summary, prior work has advanced in two main directions: preserving contextual and semantic representations of reviews, and reflecting the relative importance of information. Nonetheless, most approaches still rely on full review content, which conflates aspect-related information with contextual noise. Attention-based weighting methods can effectively highlight important cues, but do not inherently remove irrelevant tokens. Consequently, the informational density of document representations drops, and the signal-to-noise ratio decreases [11,12], degrading both learning efficiency and representation quality [13,14].

2.2. Aspect term extraction

Consumers write reviews to share experiences and opinions about products or services. Actual reviews frequently contain information unrelated to the product, such as personal anecdotes or situational details, which can weaken or misrepresent product information [9,10]. To address this issue, researchers have explored methods to identify product-related information within reviews. In natural language processing (NLP), aspects are commonly defined as opinion targets related to products, typically expressed as noun phrases [5,18].

ATE aims to accurately extract product attributes that consumers typically express in noun phrases. ATE has been adopted in some recommender-system studies to better capture user preferences. Representative approaches include topic modeling with LDA [7] and noun-phrase extraction methods based on part-of-speech tagging [18]. These approaches, however, depend on word co-occurrence patterns or fixed grammatical rules, and therefore struggle to incorporate contextual information; as a result, they may produce topics or terms unrelated to the product [5,22].

Recently, pretrained language models have improved both the accuracy and domain generalization of ATE. Transformer-based models trained on large corpora can capture vocabulary and contextual patterns across diverse domains; by modeling contextual relationships, they demonstrate strong performance. For example, Yang, Zhang and Li [27] proposed PyABSA, an ABSA framework built on pretrained language models such as BERT, RoBERTa, and DeBERTa [28]. In this framework, ATE is a core sub-task, enabling precise identification of product attributes from reviews. PyABSA's reproducible pipeline and structural flexibility support a range of various NLP tasks [29,30].

Despite these advances, ATE has seen limited application in review-based recommender systems. Earlier studies that used topic modeling or rule-based methods did not sufficiently incorporate contextual information, which led to low semantic precision and inferior recommendation performance relative to document-level representations [2,8]. To address this gap, ATRS automatically extracts explicit aspect terms from review documents using a pretrained ATE model and constructs refined user and item representations around those terms. By emphasizing high-density cues and reducing contextual noise, ATRS enables more accurate prediction of user preferences.

3. ATRS Frameworks

3.1. Overview

We propose an ATRS framework to reduce contextual noise in review-based recommendation and to increase the informational

density of learned representations. ATRS comprises two modules; the overall architecture is shown in Fig. 2. First, the ATE module takes a review document as input and extracts key aspect terms. The module tokenizes the text and encodes contextual information with a Transformer-based encoder. It then refines token-level representations through a Local Context Focus (LCF) module and identifies aspect terms using Begin, Inside, Outside (BIO) tagging. Second, the RS module embeds the extracted aspect terms with a CNN and fuses them with user and item latent vectors to build aspect-aware representations. A self-attention mechanism learns relative importance within each representation. Finally, the user and item representations are combined to predict the rating. The following sections describe each component in detail.

3.2. ATE module

We used PyABSA's ATE pipeline [27] to automatically identify meaningful aspect terms within reviews. PyABSA is an ABSA framework built on a pretrained BERT encoder; its modular, open-source design enables flexible adaptation across domains. These properties have allowed PyABSA to demonstrate strong performance in hotel management applications [30], and recommender-system contexts [29]. Our review datasets lack aspect-level annotations, and PyABSA provides models trained on domains similar to ours, including product and restaurant reviews. Therefore, we use the pretrained ATE model purely for inference without additional fine-tuning or domain adaptation, following prior work that adopts PyABSA similarly [29].

Among the ABSA backbone models in PyABSA, we adopt a DeBERTa-v3-base model. This model incorporates an LCF mechanism [31], which assigns greater weight to local semantic cues closely related to the target aspect, and it has shown strong performance on ATE benchmarks. The model processes each input sentence as follows. First, the sentence is split into subword units; token, segment, and position embeddings are combined to form initial representations. A Transformer encoder captures global context, while the LCF module emphasizes important local cues to generate contextual embeddings for each token [32]. These embeddings are passed to a BIO tagging layer, which detects the beginning and internal positions of aspect terms and extracts the corresponding words. Through this structure, PyABSA reduces contextual noise and extracts essential information from review texts. The overall ATE process and examples of extracted aspect terms are illustrated in Fig. 3.

For each review, the ATE model outputs a sequence of aspect terms, which typically consist of one or more contiguous tokens. In short or uninformative reviews (for example, “product is good”), no aspect terms may be extracted. In such cases, the review is retained and assigned an empty aspect set rather than being removed. This approach preserves the original distribution of review content, prevents additional selection bias, and ensures fair comparison with baseline models that rely on full review documents.

We group extracted aspect terms by users and by items. Define the aspect-term set for user U_i as $AT_i^U = [a_1, a_2, \dots, a_n]$ and for item V_j as $AT_j^V = [b_1, b_2, \dots, b_m]$, where a_n and b_m denote individual aspect terms, and n and m may differ. Before use in the RS module, these textual aspect terms are converted into embedding sequences (see Section 3.3) for semantic encoding. The resulting aspect representations are combined with user and item latent representations to construct features for preference prediction.

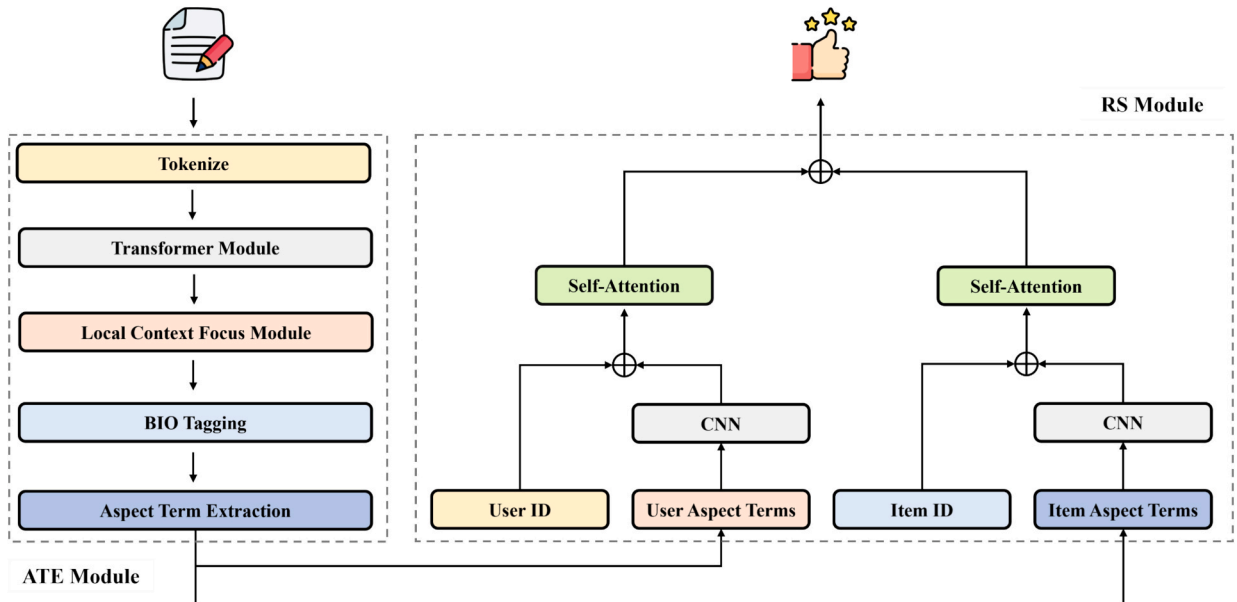


Fig. 2. ATRS Architecture.

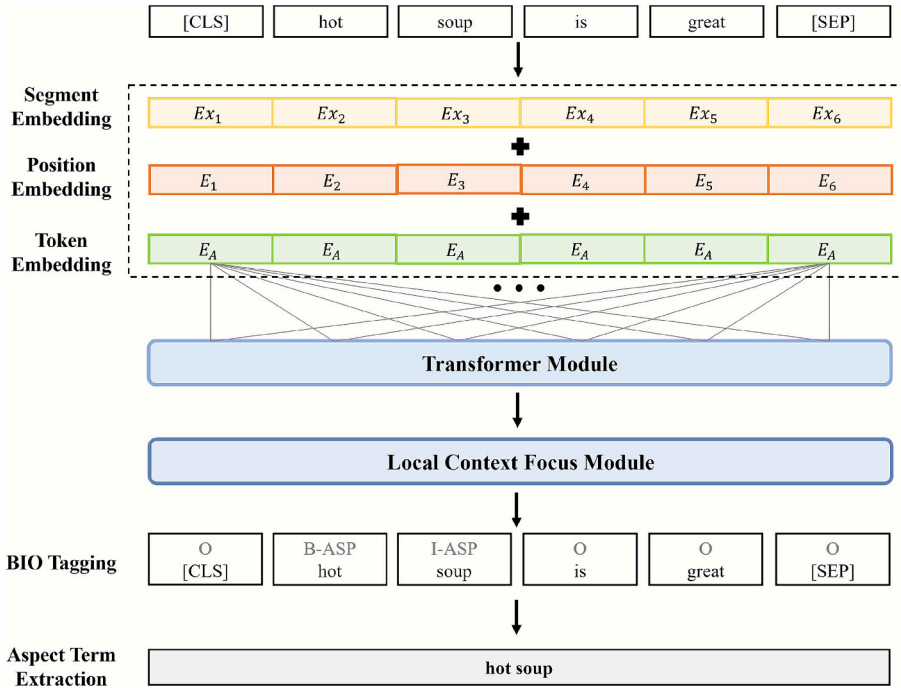


Fig. 3. Illustration of the ATE Process and its Outputs.

3.3. RS module

The RS module learns semantic representations and predicts user preferences using aspect terms extracted by the ATE module. The proposed RS module consists of four main components: (1) aspect representations learned with Word2Vec and CNN, (2) latent identity representations for users and items, (3) self-attention-based importance learning, and (4) final rating prediction. This multi-stage design incorporates high informational-density features to improve accurate preference prediction.

Aspect terms extracted for each user and item are converted into Word2Vec embedding sequences and then encoded into fixed-length vectors via convolution and pooling. These vectors serve as the aspect-term representations for users and items. In parallel, user and item identities are embedded into latent vectors. The aspect-term representations are fused with the corresponding latent identity embeddings to form aspect-aware representations. A self-attention mechanism is applied independently to these representations to model interactions and relative importance among components. Finally, user and item representations are combined to predict the rating. Detailed designs of each component are described in the following subsections. Fig. 4 illustrates the RS module architecture.

3.3.1. Aspect term semantic encoding

We apply Word2Vec embeddings [33] to transform aspect terms extracted by the ATE module for users and items into semantic vectors. Word2Vec efficiently captures semantic similarity from large corpora and preserves compound expressions such as “headphone quality” and “hot soup.” Embedding hyperparameters are: dimension = 300, window = 5, and minimum frequency = 1 to effectively include relatively sparse aspect terms [34–36]. Each extracted aspect term in AT_i^U and AT_j^V is first mapped to its Word2Vec vector, forming an ordered embedding sequence for each user and item prior to CNN processing.

The embedded aspect term sequences are fed into a one-dimensional (1-D) CNN to capture local semantic patterns. CNNs effectively model semantic patterns across adjacent tokens [37], and are well suited to emphasize informative features in short aspect-term sequences. We use a 1-D convolutional layer with 100 filters and a kernel size of 5, followed by global max pooling to select the most informative features.

As a result, the aspect term sets AT_i^U and AT_j^V are encoded into fixed-length aspect representations at_{ui} and at_{vj} for each user and item. These representations are combined with corresponding latent identity embeddings in the next stage.

3.3.2. Aspect-Aware representation Construction

We embed user and item identities into latent vectors to uniquely represent them in a learnable form. While aspect terms effectively capture user preferences and item attributes, the same terms may appear across different users or items and therefore cannot uniquely identify each entity. Incorporating identity embeddings ensures explicit differentiation and improves the accuracy of personalized preference prediction [6].

Let $U = [u_1, u_2, \dots, u_n]$ and $V = [v_1, v_2, \dots, v_m]$ denote the sets of users and items. Each identity is mapped to an integer index and

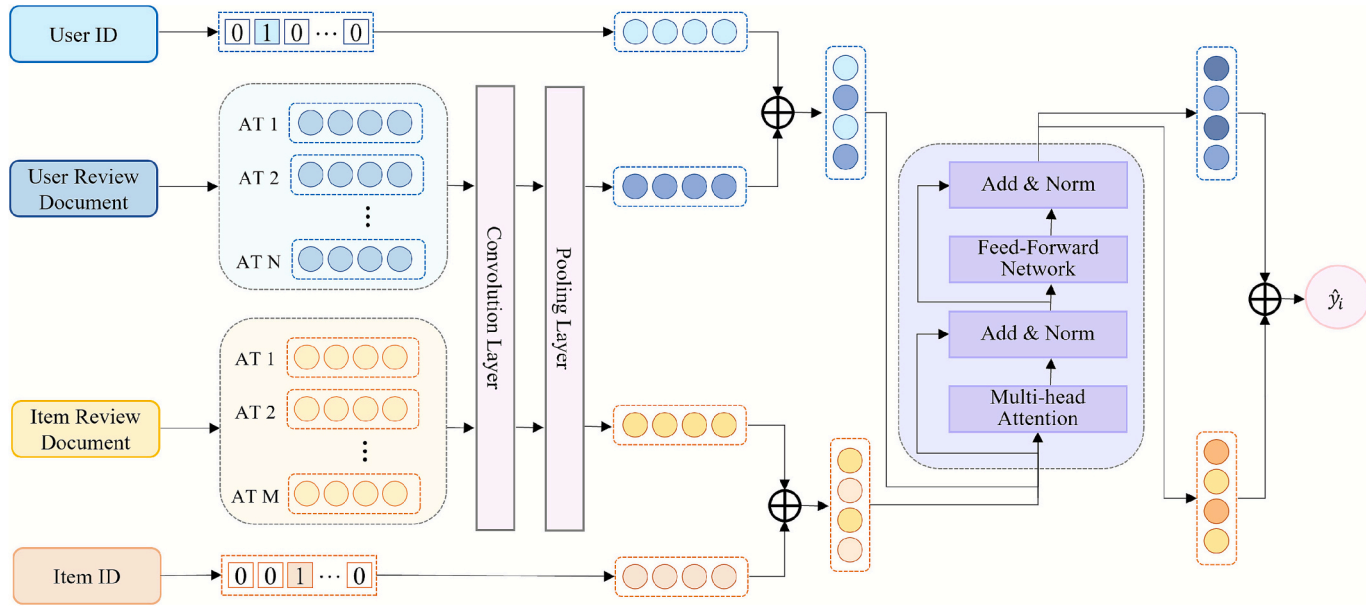


Fig. 4. RS Module.

passed through an embedding layer to produce fixed-dimensional latent vectors:

$$e_{vj} = \text{Embedding}(v_j), \quad (1)$$

$$e_{ui} = \text{Embedding}(u_i), \quad (2)$$

where e_{ui} and e_{vj} denote the latent embedding vectors of user and item, respectively. Each latent identity embedding is then combined with the corresponding aspect-term representation at_{ui} or at_{vj} to construct integrated representations. In this manuscript, we express this combination as concatenation:

$$z_{ui} = \text{Concatenate}[at_{ui}, e_{ui}], \quad (3)$$

$$z_{vj} = \text{Concatenate}[at_{vj}, e_{vj}], \quad (4)$$

where z_{ui} and z_{vj} denote the final aspect-aware representations of user and item, respectively. These representations incorporate both identity and review-based semantic features. A self-attention module then models their internal interactions dynamically.

3.3.3. Attentive Intra-representation modeling

To capture semantic interactions among components of user and item aspect-aware representations, we apply a multi-head self-attention mechanism. Aspect term representations and latent identity embeddings are heterogeneous, and simple fusion methods may fail to capture their latent interactions [38]. Self-attention mechanism dynamically models correlations and relative importance among all elements of an input sequence, enabling effective integration of heterogeneous features [39]. Therefore, in this study, the self-attention mechanism is applied to the integrated user and item representations z_{ui} and z_{vj} to strengthen semantic interactions within user and item representations and to enhance informational density. We use the standard scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \quad (5)$$

where Q , K , and V denote the query, key, and value matrices, respectively, and d_k is the dimensionality of the query and key vectors. The attention mechanism computes attention scores through the dot product between queries and keys, and normalizes them with the softmax function. The final output is then obtained as a weighted sum of the value matrix. In the self-attention mechanism, $Q = K = V$, which enables the model to learn semantic interactions among elements within the same representation. Multi-head attention performs these operations in parallel across different representation subspaces, allowing for richer representation learning. This structure is defined as follows:

$$\begin{aligned} \text{Multihead}(Q, K, V) &= \text{Concatenate}[\text{head}_1, \text{head}_2, \dots, \text{head}_N]W^O, \\ \text{where } \text{head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \end{aligned} \quad (6)$$

where W_i^Q , W_i^K , and W_i^V are trainable projection matrices for the i -th head, and W^O projects the concatenated heads. We apply multi-head self-attention independently to the integrated user and item representations:

$$U_{att} = \text{Multihead}(z_{ui}, z_{ui}, z_{ui}), \quad (7)$$

$$V_{att} = \text{Multihead}(z_{vj}, z_{vj}, z_{vj}), \quad (8)$$

where U_{att} and V_{att} denote the attention-enhanced user and item representations. Following the Transformer pattern, these representations are further refined through a feed-forward network with nonlinear transformations. To reduce information loss and stabilize training, residual connections and layer normalization are applied [39]. This process is defined as follows:

$$F_U = \text{LN}(\text{FFN}(U_{att}) + U_{att}), \quad (9)$$

$$F_V = \text{LN}(\text{FFN}(V_{att}) + V_{att}), \quad (10)$$

where F_U and F_V denote the final refined user and item representations, respectively. These refined representations are used for preference prediction.

3.3.4. Rating prediction

In this stage, the final refined representations of users and items are combined to predict ratings. First, the user representation F_U and the item representation F_V are concatenated to preserve information from both representations. This process is defined as follows:

$$O = \text{Concatenate}[F_U, F_V], \quad (11)$$

where O denotes the combined representation of the user and the item. The resulting vector is then fed into a multilayer perceptron (MLP) to model high-dimensional interaction features between users and items. The MLP consists of multiple hidden layers with

nonlinear activation functions, and the final output is the predicted rating $\hat{y}$ for a given user–item pair. This process is formulated as follows:

$$\begin{aligned} FF_1 &= \text{ReLU}(W_1 \cdot O + b_1), \\ &\dots \\ \hat{y} &= \text{linear}(W_n \cdot FF_{n-1} + b_n). \end{aligned} \quad (12)$$

where W_1 and b_1 denote the weight matrix and bias of the first hidden layer, respectively, and W_n and b_n denote those of the final prediction layer. FF_{n-1} represents the output of the last hidden layer. The model is trained by minimizing the difference between the predicted rating $\hat{y}$ and the true rating y using the mean squared error (MSE) loss function.

4. Experiments

4.1. Datasets

To evaluate the performance of the proposed ATRS framework, this study uses three datasets: Amazon Musical Instruments, Amazon Video Games, and Yelp Restaurant. These datasets span different domains and scales, allowing the generalization capability of the model to be examined [8]. Amazon, the world's largest e-commerce platform, provides large-scale review data across diverse product categories [40]. Yelp is a global review platform that provides reviews and business information in the food and beverage domain and is widely used in recommender system research [41]. The Yelp dataset contains sparse reviews across regions [8]; therefore, data from Pennsylvania, which has the largest number of reviews, is used to reduce regional variance. From each dataset, user and item identities, review texts, and ratings are extracted as model inputs. All ratings range from 1 to 5, where 5 indicates the highest preference and 1 the lowest.

This study followed preprocessing strategies to ensure accurate preference prediction. First, the Amazon datasets include reviews written without verified purchases; to retain only authentic reviews, those from unverified users were removed [42]. Similarly, for the Yelp dataset, only reviews associated with restaurants and currently operating businesses were retained. Second, reviews without textual content or missing ratings were removed. Third, to ensure training stability, only users and items with at least five reviews were retained [8]. Fourth, text preprocessing was applied to improve aspect term extraction. Non-English reviews, URLs, HTML tags, and stopwords were removed [40]. In addition, all texts were lowercased, abbreviations were restored, and lemmatization was applied to avoid treating semantically equivalent words as distinct terms [1]. After preprocessing, reviews containing missing values were removed to maintain data integrity.

The statistics of the final datasets are summarized in Table 1. The datasets differ in scale, average review length, and skewness across domains. Longer reviews may include more contextual or peripheral information, which can potentially lower informational density [43]. Skewness reflects variation in review length, with higher values indicating that some reviews are substantially longer than others. The data were randomly split into training, validation, and test sets with ratios of 70 %, 10 %, and 20 %, respectively. All experiments were conducted using datasets processed under identical preprocessing criteria.

4.2. Evaluation metrics

To quantitatively evaluate model performance, four metrics widely used in recommender system research were employed: mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and mean absolute percentage error (MAPE) [44,45].

MAE measures the average absolute difference between predicted and actual values and provides an intuitive estimate of prediction accuracy. MSE computes the average of squared errors, assigning greater penalties to larger deviations and thus being more sensitive to outliers. RMSE, the square root of MSE, preserves the original rating scale while remaining sensitive to significant errors. MAPE expresses prediction error as a percentage of the actual values, enabling evaluation from a relative perspective. The formulations of these metrics are given as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (13)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (14)$$

Table 1

Descriptive statistics of the amazon and yelp dataset.

Dataset	Number of Reviews	Number of Users	Number of Items	Review Length	Skewness
Musical Instruments	449,773	51,655	21,140	245.984	6.535
Video Games	577,488	71,755	19,202	273.021	8.959
Yelp	479,235	34,085	7,218	618.529	2.385

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (15)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100. \quad (16)$$

By employing these complementary metrics, model performance is evaluated in terms of accuracy, robustness, and sensitivity to outliers.

4.3. Baselines

To empirically validate the effectiveness of the proposed ATRS framework, it was compared with seven representative recommender system baselines. The baselines are categorized into rating-based and review-based models according to their input information, each with distinct structural characteristics.

- PMF [46] is a probabilistic matrix factorization approach that decomposes the user–item rating matrix into low-dimensional latent factor matrices.
- NCF [4] is a neural collaborative filtering model that learns latent user and item representations from rating data and combines generalized MF (GMF) with MLPs to capture complex nonlinear interactions.
- DeepCoNN [2] is a review-based model that aggregates all reviews written by a user or received by an item into documents and employs parallel CNNs to learn contextual representations for modeling semantic user–item interactions.
- NARRE [8] assigns higher weights to informative reviews using review-level attention over CNN-based embeddings, reflecting differences in the importance of individual reviews.
- AENAR [6] learns representations using CNNs and applies aspect-level attention to emphasize the most informative aspects within the entire review text.
- SAFMR [26] refines review representations by applying self-attention to CNN-based embeddings, capturing feature dependencies and relative importance within reviews.
- MFNR [23] is a multi-embedding fusion model that combines representations from multiple pretrained language models, such as BERT and RoBERTa, to enhance the diversity and precision of review representations..

Table 2
Predictive performance of the proposed model and baselines.

Dataset	Factor	Model	MAE	MSE	RMSE	MAPE
Musical Instruments	Rating-based	PMF	1.306	2.640	1.625	35.034
		NCF	1.174	1.705	1.306	35.401
	Review-based	DeepCoNN	0.786	1.137	1.067	29.931
		NARRE	0.767	0.993	0.997	29.459
		AENAR	<u>0.665</u>	0.970	0.985	27.193
		SAFMR	0.705	0.975	0.987	28.388
		MFNR	0.708	<u>0.965</u>	<u>0.982</u>	<u>26.922</u>
		ATRS	0.640	0.933	0.966	26.638
Video Games	Rating-based	PMF	1.220	2.407	1.551	33.948
		NCF	0.948	1.331	1.154	35.032
	Review-based	DeepCoNN	0.847	1.263	1.124	32.850
		NARRE	0.776	1.173	1.083	30.518
		AENAR	<u>0.693</u>	1.002	1.001	28.039
		SAFMR	0.711	1.033	1.016	30.016
		MFNR	0.730	<u>0.980</u>	<u>0.990</u>	<u>27.863</u>
		ATRS	0.646	0.970	0.985	27.537
Yelp	Rating-based	PMF	1.276	2.803	1.674	38.330
		NCF	1.085	1.674	1.294	39.320
	Review-based	DeepCoNN	0.937	1.381	1.175	38.276
		NARRE	0.886	1.212	1.101	36.724
		AENAR	<u>0.845</u>	1.177	1.085	35.605
		SAFMR	0.881	1.229	1.109	36.076
		MFNR	0.855	<u>1.174</u>	<u>1.084</u>	33.923
		ATRS	0.832	1.163	1.078	<u>34.917</u>

* Boldface indicates best performance; underlining indicates the second-best performance.

4.4. Experiment settings

To optimize the performance of ATRS, repeated experiments were conducted to tune key hyperparameters. The number of attention heads was searched in the range [4, 6, 8, 10, 12], the batch size in [64, 128, 256, 512], and the learning rate in [0.001, 0.002, 0.005, 0.0001, 0.0002, 0.0005]. The dropout rate, used to prevent overfitting, was explored in the range [0.1, 0.2, 0.3, 0.4, 0.5]. The dimensionality of the latent feature was tuned in [32, 64, 128, 256]. Based on these searches, the final hyperparameter settings were 12 attention heads, a batch size of 128, a learning rate of 0.001, a dropout rate of 0.1, and a latent dimension of 128.

Model training employed the Adam (Adaptive Moment Estimation) optimizer. To prevent overfitting, early stopping with a patience of five epochs without improvement in validation loss was applied [1,40]. To reduce randomness and improve reliability, each experiment was repeated five times, and the average performance was reported. The parameters of baseline models were set according to the ranges recommended in their original studies, and the best-performing configurations were selected. All experiments were conducted on a cloud server equipped with an NVIDIA H100 GPU, 128 GB RAM, and a virtual environment based on Python 3.10 and TensorFlow 2.15.

5. Results and discussion

5.1. Performance comparison with baseline models

We evaluated the effectiveness of the proposed ATRS against the baselines described in Section 4.3. Results are summarized in Table 2. ATRS achieved the best performance across all datasets, with average improvements of 19.54 % in MAE, 24.02 % in MSE, 11.89 % in RMSE, and 8.41 % in MAPE.

Across datasets, rating-based models (PMF and NCF) produced higher errors than review-based approaches on all metrics. PMF relies solely on matrix factorization of the user–item rating matrix and is therefore sensitive to data sparsity, which yields the lowest overall performance. Although NCF models nonlinear user–item interactions, it depends only on user and item identifiers and thus cannot capture fine-grained textual preferences.

Review-based models show consistent improvements by incorporating textual information. DeepCoNN benefits from contextual encodings but cannot distinguish the importance of individual reviews. NARRE applies review-level attention to address that limitation and reduces prediction error. AENAR and SAFMR model aspect-level or contextual interactions via attention mechanisms and achieve further gains. MFNR, which fuses embeddings from pretrained language models, performs strongly among baseline models, reflecting the value of rich semantic representations.

ATRS consistently attains superior performance across datasets and metrics. It records the lowest MAE, MSE, and RMSE on both Amazon datasets. Compared with MFNR, ATRS yields slightly higher (worse) MAPE on Yelp, suggesting that aspect-term representations may be less stable when reviews are substantially longer and more skewed. Overall, ATRS benefits from explicitly extracting aspect terms and constructing user and item representations from them. By filtering contextual noise and focusing on informative cues, ATRS increases the informational density of learned representations and improves prediction accuracy.

From these results, we draw three main implications. First, review texts convey user preferences that ratings alone cannot fully capture. Second, full reviews include contextual noise that can limit performance when exploited indiscriminately. Third, selective use of aspect-level information raises informational density and the signal-to-noise ratio, enabling more accurate representation learning and improved recommendation accuracy.

5.2. Impact of aspect term number on prediction performance

ATRS relies on aspect terms extracted from user and item reviews as its primary input. We analyzed how the number of aspect terms affects prediction performance. For each user and item, extracted aspect terms were ranked by frequency, and subsets containing 25 %, 50 %, 75 %, 90 %, and 100 % of the terms were evaluated. The number of aspect terms at each ratio appears in Table 3; performance results are summarized in Table 4. The results show that the optimal number of aspect terms varies across domains.

In product domains (Musical Instruments, Video Games), restricting aspect terms to 50 % or fewer removed informative signals and degraded overall performance. Using 75 % or more produced improved metrics, with the best results near 90 %–100 %. In the restaurant domain (Yelp), the best performance occurred at 75 % and declined when more than 90 % of terms were included. Two explanations are likely. First, semantically similar expressions increase redundancy and overlap, which reduces learning efficiency.

Table 3

Number of aspect terms for users and items across the dataset.

Dataset	User/Item	25 %	50 %	75 %	90 %	100 %
Musical Instruments	User	9	14	25	45	386
	Item	23	64	197	602	4,490
Video Games	User	2	8	15	25	243
	Item	26	78	211	477	4,236
Yelp	User	26	56	145	312	3,206
	Item	186	441	1,038	2,079	5,831

Table 4

Performance comparison across different numbers of aspect terms.

Dataset	Metrics	25 %	50 %	75 %	90 %	100 %
Musical Instruments	MAE	0.662	0.694	0.643	0.640	<u>0.642</u>
	MSE	0.944	0.945	0.934	<u>0.933</u>	0.932
	RMSE	0.971	0.972	0.967	<u>0.966</u>	0.966
	MAPE	26.903	27.551	26.841	26.638	<u>26.649</u>
Video Games	MAE	0.659	0.658	0.649	<u>0.647</u>	0.646
	MSE	0.989	0.974	0.974	<u>0.973</u>	0.970
	RMSE	0.995	<u>0.987</u>	<u>0.987</u>	<u>0.987</u>	0.985
	MAPE	27.900	27.750	<u>27.640</u>	27.900	27.537
Yelp	MAE	<u>0.838</u>	0.840	0.832	0.842	0.856
	MSE	<u>1.165</u>	1.167	1.163	1.170	1.173
	RMSE	<u>1.080</u>	<u>1.080</u>	1.078	1.082	1.083
	MAPE	35.438	35.405	34.917	<u>34.920</u>	35.672

* Boldface indicates best performance; underlining indicates the second-best performance.

Second, because Word2Vec used a minimum frequency threshold of 1, rare terms are included; these low-frequency terms can introduce noise and degrade prediction performance.

Domain characteristics further explain these differences. Amazon reviews often describe objective, product-oriented attributes (for example, “*clip tuner*,” “*HDMI processing*,” “*battery backup*”), focusing on functions, performance, and design [47], which are relatively consistent and repetitive; including many aspect terms therefore preserves useful information. In contrast, Yelp reviews emphasize subjective, service-oriented experiences (for example, “*romantic ambience*,” “*community feel*,” “*restaurant design*”) [48], where diverse expression styles increase term variability and redundancy. Consequently, excessive aspect terms can reduce model performance in service-oriented domains.

Overall, this experiment shows that the number of aspect terms has a significant impact on model performance, and that optimizing informational density benefits from domain-specific strategies: include more terms in product-oriented domains, but be selective in service-oriented domains.

5.3. Ablation study

We performed ablation experiments to quantify the contribution of ATRS components. The analysis is divided into architecture-level and input-representation-level ablations.

5.3.1. Ablation on model architecture

This subsection evaluates the contribution of the self-attention module. ATRS integrates identity embeddings, representing stable user and item interaction patterns, with aspect-term embeddings, encoding fine-grained semantic cues. Because these features differ semantically, simple concatenation may fail to capture latent relationships [38,39]. We compare models with and without the self-attention module. Results in Table 5 show that adding self-attention consistently improves performance.

The improvement stems from self-attention dynamic weighting [39]; it assigns greater weight to contextually relevant features while suppressing noise, which raises informational density of user and item representations. Even when the same aspect term appears across entities, its contribution varies by user or item, enabling more personalized representations. Multi-head attention captures semantic relations from multiple perspectives, increasing expressiveness and yielding consistent accuracy gains.

Table 5

Effectiveness of self-attention for aspect-aware representation.

Dataset	Self-Attention	MAE	MSE	RMSE	MAPE
Musical Instruments	w/o	0.660	0.948	0.974	27.535
	With	0.640	0.933	0.966	26.638
Video Games	w/o	0.701	0.984	0.992	28.738
	With	0.646	0.970	0.985	27.537
Yelp	w/o	0.851	1.174	1.083	35.340
	With	0.832	1.163	1.078	34.917

* Boldface indicates the best performance.

5.3.2. Ablation on input representation

This subsection examines the effect of aspect-term extraction in ATRS from the input representation perspective. Real-world reviews typically contain product-related information and contextual noise, which reduces informational density and makes accurate preference prediction more difficult [9,10]. To address this issue, ATRS constructs user and item documents from extracted aspect terms rather than full review texts. We compare ATRS with an alternative, ATRS-A, that uses full review documents as input. Experimental results show that ATRS consistently outperforms ATRS-A. Detailed results are reported in Table 6.

This performance gap indicates that the informational density of the input representation plays a critical role in recommendation accuracy. ATRS and ATRS-A follow the same architecture, applying Word2Vec embeddings and CNN-based encoding to the input documents to extract semantic information and form unified representations. However, ATRS-A's full-text inputs contain more contextual noise, resulting in lower informational density and making it harder for the self-attention mechanism to weight informative signals accurately. Explicitly extracting aspect terms focuses representation learning on preference-relevant cues and improves recommendation performance.

5.4. Case study

Explainability in recommender systems is often assessed qualitatively [49,50]. We present an illustrative case of the aspect-term recommendation process (Fig. 5). The user's past reviews for Nintendo-related products (for example, Nintendo Switch Console, Wii Controllers, Sonic Superstars, Animal Crossing, Super Mario Party) yield extracted aspect terms such as family, multiplayer, smooth control, competition, and kid-friendly. These terms indicate a preference for family-oriented play, motion-based interactions, easy control, and casual competitive entertainment.

For a candidate item, Nintendo Switch Sports, extracted aspect terms include family play, motion control, easy control, competition, and entertainment, showing strong semantic alignment with the user's preferences. This alignment explains the predicted rating of 4.83 and demonstrates that ATRS grounds predictions in interpretable aspect-level information. The case illustrates how the ATRS framework effectively reduces contextual noise from review text and retains meaningful aspect terms that characterize user preferences, improving transparency and trustworthiness.

6. Conclusion and future work

This study proposed an ATRS framework to reduce contextual noise in review-based recommendation and to increase the informational density of learned representations. Unlike document-level approaches that include irrelevant personal or contextual details, ATRS uses an ATE module to extract key aspect terms that reflect user preferences and item attributes. These terms are encoded by a CNN and integrated with identity embeddings via self-attention to form high-density representations. Experiments on Amazon and Yelp datasets show that ATRS consistently outperforms baseline models. The analysis further indicates that the effect of aspect-term quantity varies by domain, reflecting differences in informational density and structural characteristics of user representations. Overall, this study shows that selective incorporation of essential information and targeted representation refinement substantially improve recommendation accuracy, underscoring the value of aspect-based representation learning as a guiding principle for the design of future recommender systems.

This work has several limitations that point to future research directions. First, the model is trained on reviews collected at a single time point and therefore does not capture temporal dynamics such as shifting user preferences or product updates. Future work could incorporate temporal modeling to reflect evolving behavior. Second, this study focuses solely on textual information, while real-world reviews often include images or video that convey visual attributes and usage contexts; a multimodal extension that integrates text and visual information would likely provide a richer representation of user preferences. Future work could explore multimodal recommender systems that jointly learn from textual and visual information to fully exploit this potential. Third, although a BERT-based ATE model extracts aspect terms, the semantic contribution of specific terms to predictions is not fully interpretable. Future work could explore generative-model-based explanations or natural-language explanations to improve transparency and interpretability. Finally, domain-adaptive ATE fine-tuning and strategies for handling low-frequency aspect terms merit further investigation.

In conclusion, ATRS demonstrates that emphasizing high-informational-density cues in reviews and refining representations around aspect-level terms improves recommendation accuracy. These findings encourage future work on targeted representation learning and explainable recommendations.

CRedit authorship contribution statement

Heena Lim: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Xinzhe Li:** Software, Methodology, Formal analysis, Data curation. **Seonu Park:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Qinglong Li:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Jaekyeong Kim:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

Table 6
Performance comparison of ATRS and ATRS-A.

Dataset	Model	MAE	MSE	RMSE	MAPE
Musical Instruments	ATRS-A	0.675	0.961	0.980	27.444
	ATRS	0.640	0.933	0.966	26.638
Video Games	ATRS-A	0.693	0.990	0.995	28.439
	ATRS	0.647	0.970	0.985	27.537
Yelp	ATRS-A	0.843	1.184	1.088	35.817
	ATRS	0.832	1.163	1.078	34.917

* Boldface indicates the best performance.



Fig. 5. Aspect-term based Recommendation Process for a Target Item.

influence the work reported in this paper.

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Data availability

Data will be made available on request.

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